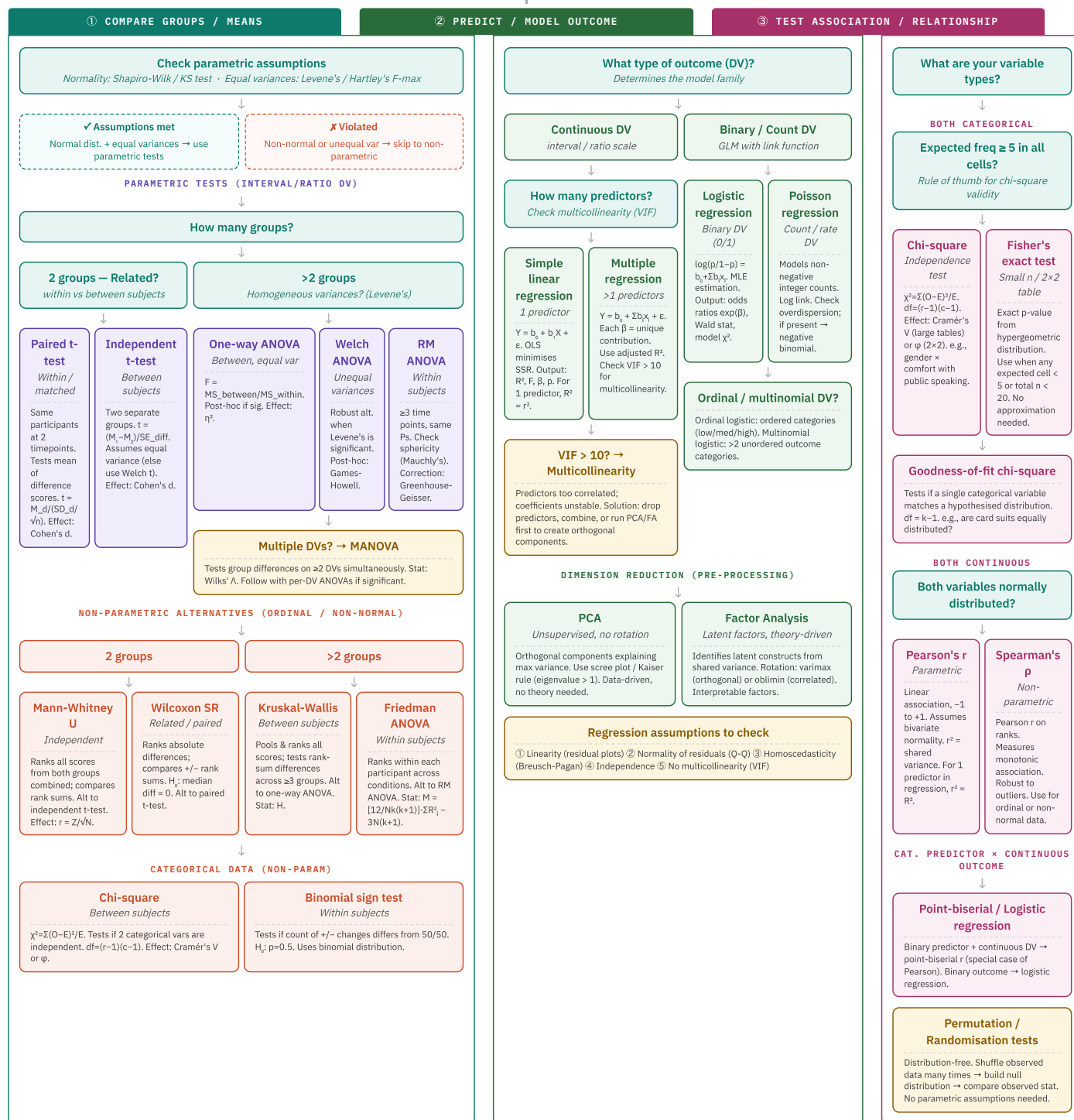


Statistical Test Selection — Flowchart & Reference

Sources: BRSM Lectures · Hypothesis Testing · ANOVA · Non-parametric Tests · GLM · Regression · Bayesian Statistics

PAGE 1 OF 3 — DECISION FLOWCHART (WITH MICRO-DEFINITIONS)

START — What is your research goal?



Multiple tests? → Apply corrections

k tests at $\alpha=0.05$ inflates error to $1-(0.95)^k$. Bonferroni: α/k (conservative). FDR/Benjamini-Hochberg: controls false discovery rate (less conservative). Holm-Bonferroni: step-down compromise.

Bayesian alternative (any test)

Prior + likelihood → posterior. Bayes Factor BF_{10} : >3 moderate evidence for H_1 , >10 strong. Reports credible intervals (CrI) not p-values. Bayesian paired t-test replaces frequentist equivalent.

All tests share the same logic

Define H_0/H_1 → Set α (0.05) → Compute test statistic → Get p-value → Report effect size (d , η^2 , r , R^2) → Check power ($1-\beta$)

LEGEND

Decision / question Parametric test Non-parametric test Regression / GLM Categorical / association Caution / correction Bayesian framework

PARAMETRIC TESTS	
Independent t-testPARAMETRIC DEFINITION Tests whether the means of two independent, unrelated groups differ significantly. Assumes normality within each group and homogeneity of variance. WHEN TO USE Two separate groups; continuous interval/ratio DV; parametric assumptions met. FORMULA $t = (M_1 - M_2) / SE_{diff} \quad \cdot \quad df = n_1 + n_2 - 2$ EFFECT SIZE Cohen's d = (M₁ - M₂) / SD_{pooled}. Trivial <0.1, small 0.1–0.3, moderate 0.3–0.5, large >0.5. Example: Do male and female students differ in exam scores? IV = gender (2 levels), DV = score (continuous).	Paired (Related) t-testPARAMETRIC DEFINITION Compares means from the same participants at two timepoints, or from matched pairs. Operates on the difference scores d = X₁ - X₂. WHEN TO USE Within-subjects or matched pairs design. Normality of difference scores required. FORMULA $t = M_d / (SD_d / \sqrt{n}) \quad \cdot \quad df = n - 1$ EFFECT SIZE Cohen's d = M_d / SD_d. Example: Do knowledge scores improve after a group discussion? Pre-test M=83, post-test M=84, n=1000. Significant statistically but d is tiny — always report effect size!
One-way ANOVAPARAMETRIC DEFINITION Tests whether at least one group mean differs among ≥3 independent groups. Partitions total variance into between-group (SS_b) and within-group (SS_w) components. ASSUMPTIONS Normality (Shapiro-Wilk); homogeneity of variance (Levene's). Robust for n > 25/group. If largest variance < 4–5× smallest, ANOVA results remain valid. FORMULA $F = MS_{between} / MS_{within} \quad \cdot \quad df_b = G-1, df_w = N-G$ POST-HOC Only if F is significant: Tukey HSD, Bonferroni, or Games-Howell (unequal n/var). EFFECT SIZE η² = SS_{between} / SS_{total}. Small: 0.01, medium: 0.06, large: 0.14. Example: Do three teaching methods (lecture, flipped, online) produce different exam averages? IV = method (3 levels), DV = score.	Repeated Measures ANOVAPARAMETRIC DEFINITION ANOVA for within-subjects designs: same participants measured ≥3 times. Removes individual differences from error variance → higher power than between-subjects ANOVA. KEY ASSUMPTION Sphericity — variances of all pairwise difference scores must be equal. Test: Mauchly's W. If violated: Greenhouse-Geisser or Huynh-Feldt epsilon correction. EFFECT SIZE Partial η² or generalised η². Example: Anxiety scores at baseline, 4 weeks, and 8 weeks of therapy — same participants. Check sphericity before reporting F.
Welch ANOVAPARAMETRIC DEFINITION Robust alternative to one-way ANOVA that does not assume equal variances. Uses weighted group means. Recommended when Levene's test is significant. POST-HOC Games-Howell: does not assume equal variances or equal sample sizes. Example: Comparing salaries across 4 departments where one department has much higher variance than others → Welch ANOVA + Games-Howell.	MANOVAPARAMETRIC DEFINITION Multivariate ANOVA: tests group differences on ≥2 DVs simultaneously. Controls family-wise Type I error across DVs. Requires multivariate normality. STATISTICS Wilks' Λ (most common), Pillai's trace (most robust), Hotelling's T², Roy's largest root. FOLLOW-UP If MANOVA is significant, run individual ANOVAs per DV with Bonferroni correction. Example: Do three therapy types differ on both anxiety AND depression simultaneously? One MANOVA instead of two separate ANOVAs.
NON-PARAMETRIC TESTS	
Mann-Whitney U TestNON-PARAM DEFINITION Non-parametric equivalent of the independent t-test. Combines and ranks all scores from both groups; tests whether one group's ranks are systematically higher (i.e., different medians). Between-subjects design, skewed distributions, ordinal data. ASSUMPTION A real difference between two populations should cause one sample's scores to be generally larger. If combined and ranked, larger ranks concentrate in one group. H₀ The two populations share an identical distribution (same median). EFFECT SIZE $r = Z / \sqrt{N}$ Example: Pain ratings (1–10 Likert, skewed) for Drug A vs Drug B. Non-normal → Mann-Whitney U instead of independent t-test.	Wilcoxon Signed-Rank TestNON-PARAM DEFINITION Non-parametric equivalent of the paired t-test. Computes difference scores, ranks their absolute values, then compares the sum of positive vs negative ranks. Within-subjects or matched pairs design. H₀ Population median difference = 0. Reject H₀ if observed W ≤ critical W value. DATA LEVEL Ordinal (rank order). Also valid for discrete related samples where parametric assumptions are violated. Example: Did students perform better in mock exam vs final GRE? H₀: med diff = 0; H₁: med diff > 0 (1-tailed). If paired differences are non-normal → Wilcoxon SR.
Kruskal-Wallis TestNON-PARAM DEFINITION Non-parametric equivalent of one-way ANOVA. All scores from ≥3 independent groups are pooled and ranked; tests if rank sums differ significantly between groups. WHEN TO USE ≥3 independent groups; ordinal DV or violated normality; use when ANOVA assumptions fail. FORMULA $H = [12 / N(N+1)] \times \sum (R_k^2 / n_k) - 3(N+1)$ Example: Job satisfaction (ordinal 1–5) across 3 departments. Non-normal data → Kruskal-Wallis (H statistic) instead of one-way ANOVA.	Friedman's ANOVANON-PARAM DEFINITION Non-parametric equivalent of repeated measures ANOVA. Ranks scores within each participant (block) across ≥3 conditions, then compares rank sums across conditions. FORMULA $M = [12 / Nk(k+1)] \times \sum R_i^2 - 3N(k+1)$ WHERE k = number of conditions, N = number of participants, R_i = sum of ranks for condition i. Example: 6 students rate 3 study strategies. Ranks within each student: Reread=1, Answer Qs=2, Create Qs=3. Friedman M=9.33, df=2, p=.009 → strategies differ significantly.
Chi-square IndependenceCATEGORICAL DEFINITION	Binomial Sign TestCATEGORICAL DEFINITION

Tests whether two categorical variables are independent. Compares observed frequencies (O) to expected frequencies (E) assuming independence.

ASSUMPTIONS
Expected frequency ≥ 5 in each cell. $df = (rows-1)(cols-1)$.

FORMULA

$$\chi^2 = \sum (O - E)^2 / E \quad \cdot \quad E = (row\ total \times col\ total) / N$$

EFFECT SIZE
2x2: $\phi = \sqrt{\chi^2/n}$. Larger: Cramér's V = $\sqrt{(\chi^2/n \cdot df^*)}$. Small=0.1, medium=0.3, large=0.5.

Example: Extroverts vs introverts — comfortable with public speaking? Extroverts: 10 comfortable / 40 not. Introverts: 40/10. $\chi^2(2, n=150)=8.23, p<.05, V=0.23 \rightarrow$ moderate effect.

Tests whether the proportion of positive change signs in a within-subjects design differs from 50% (chance). Uses the binomial distribution. The simplest non-parametric test for related pairs.

H_a
 $p(\text{positive change}) = 0.5$. Count positives (S); compare to binomial critical value.

WHEN TO USE
Only sign of change is available (not magnitude). Related pairs or matched design.

Example: After diet intervention, 18 of 20 participants lost weight (+). $p(S \geq 18 \mid p=0.5) = 0.0002 \rightarrow$ reject H₀, diet had a significant positive effect.

REGRESSION & GLM

Simple Linear RegressionREGRESSION

DEFINITION

Models the linear relationship between one continuous predictor (X) and one continuous outcome (Y). Uses Ordinary Least Squares (OLS) to find the line minimising the sum of squared residuals (SSR).

MODEL

$$Y = b_0 + b_1X + \epsilon$$

KEY OUTPUT

R² (proportion of variance explained), F-statistic (overall model), β₁ (slope = change in Y per 1-unit increase in X), p-values. Note: for 1 predictor, R² = r² (Pearson).

ASSUMPTIONS

Linearity, normality of residuals, homoscedasticity, independence of errors. Check with residual plots and Q-Q plots.

Example (housing.csv): Predict median_house_value from median_income alone. β₁ = slope (\$ per unit income); R² = how much income explains price variation.

Multiple Linear RegressionREGRESSION

DEFINITION

Extends simple regression to ≥2 predictors. Each β = unique contribution of that predictor holding all others constant (partial regression coefficient). Adjusted R² penalises for adding extra predictors.

MODEL

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_nX_n + \epsilon$$

MULTICOLLINEARITY

VIF > 10 (or > 5 strict): predictors too correlated; coefficients become unstable and uninterpretable. Solutions: remove redundant predictor, combine them, or run PCA/FA first to create orthogonal components.

Example (housing.csv): Predict house value from income, housing_median_age, total_rooms, population, and ocean_proximity simultaneously. Check each predictor's unique contribution via standardised β.

Logistic RegressionGLM

DEFINITION

Models the probability of a binary outcome (0/1) using a logistic link function (sigmoid). Estimates coefficients via Maximum Likelihood Estimation (MLE), not OLS. Coefficients are log-odds.

MODEL

$$\log(p / 1-p) = b_0 + b_1X_1 + \dots + b_nX_n$$

INTERPRETATION

exp(β) = odds ratio. OR > 1: predictor increases odds of outcome. 1-unit increase in X multiplies odds by exp(β). Model significance: χ² likelihood ratio test.

OUTPUT

Odds ratios, Wald statistic per predictor, Nagelkerke R², classification accuracy (confusion matrix), AUC-ROC.

Example (binary.csv): Predict admission (0/1) from GRE, GPA, and rank. Each 1-point GRE increase multiplies odds of admission by exp(β_gre). Higher GPA and lower rank → higher admission probability.

PCA & Factor AnalysisDIMENSION REDUCTION

PCA

Transforms correlated variables into orthogonal (uncorrelated) principal components ordered by variance explained. Data-driven. No theory assumed. Use scree plot or Kaiser rule (eigenvalue > 1) to select components. Components are linear combinations of original variables.

FACTOR ANALYSIS

Identifies latent factors explaining shared (common) variance among observed variables. Theory-driven — you hypothesise what the factors represent. Rotation improves interpretability: varimax (orthogonal, uncorrelated factors) or oblimin (correlated factors allowed).

KEY DIFFERENCE

PCA explains total variance; FA explains only common variance. PCA = preprocessing tool for dimensionality reduction. FA = measurement model for latent constructs (e.g., psychological scales).

Example: 20-item personality questionnaire → FA reveals 5 latent factors (openness, conscientiousness...). Use factor scores as DVs in ANOVA. Alternatively, PCA on housing.csv variables to reduce multicollinearity before regression.

ASSOCIATION TESTS

Pearson's rPARAMETRIC

DEFINITION

Measures strength and direction of the linear relationship between two continuous variables. Ranges -1 to +1. 0 = no linear relationship. Association only — does not imply causation.

ASSUMPTIONS

Both variables approximately normally distributed; linear relationship (check scatterplot — Anscombe's quartet!); no extreme outliers.

FORMULA

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{[(n-1) \cdot SD_x \cdot SD_y]}$$

EFFECT SIZES

|r|: small=0.1, medium=0.3, large=0.5. r² = proportion of shared variance. For 1-predictor regression: r² = R².

Example: Is there a linear relationship between hours studied and exam score? r = 0.62 → moderate-large positive relationship. r² = 0.38 → studying explains 38% of score variance.

Spearman's ρNON-PARAMETRIC

DEFINITION

Non-parametric equivalent of Pearson's r. Computes Pearson correlation on rank-transformed data. Measures the strength of monotonic (not necessarily linear) association. Robust to outliers.

WHEN TO USE

Non-normal distributions, ordinal variables, or when outliers distort Pearson r. Use if scatterplot shows monotonic but non-linear trend.

FORMULA

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$
 (d = rank difference)

Example: Class rank (ordinal) vs anxiety level (ordinal Likert). Both ordinal → Spearman's ρ. Or: latitude vs temperature — skewed distribution → Spearman more appropriate than Pearson.

Fisher's Exact TestCATEGORICAL

DEFINITION

Calculates the exact probability of observing the given 2×2 contingency table (or more extreme tables) given fixed marginal totals, using the hypergeometric distribution. Valid for any sample size — no approximation.

WHEN TO USE

2×2 tables where expected cell frequency < 5 in any cell, or total n < 20. Use instead of chi-square when its assumptions are violated.

H₀

The two categorical variables are independent (same H₀ as chi-square).

Example: 12 patients: 3 on Drug A (2 recovered, 1 not), 9 on Drug B (3 recovered, 6 not). Small n → chi-square assumptions violated → Fisher's exact p = 0.54 → no significant difference.

Chi-square Goodness-of-fitCATEGORICAL

DEFINITION

Tests whether a single categorical variable's observed frequencies match a hypothesised (expected) distribution. Unlike the independence test, only one variable is involved.

FORMULA

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$
 df = k - 1

Example: Are 100 cards dealt equally across 4 suits (expected: 25 each)? Observed: ♠20, ♥30, ♦25, ♣25. χ²(3) = (25+25+0+0)/25 = 2.0, p = .57 → no significant deviation from equal distribution.

CORRECTIONS, BAYESIAN INFERENCE & HYPOTHESIS TESTING LOGIC

Multiple Comparisons CorrectionsCAUTION

PROBLEM

Running k tests at α=0.05 inflates family-wise Type I error to 1-(0.95)^k. At k=14 tests this exceeds 50% — you're more likely than not to get at least one false positive by chance alone.

BONFERRONI

α_{corrected} = α / k. Very conservative. Best when k is small and all comparisons are planned. e.g., 20 tests → threshold p < 0.0025 per test.

HOLM-BONFERRONI

Bayesian InferenceFRAMEWORK

DEFINITION

Updates prior beliefs with observed data (likelihood) to produce a posterior distribution over parameters. All conclusions are probabilistic statements about parameters, not about data given a null.

FORMULA

$$P(\theta \mid \text{data}) \propto P(\text{data} \mid \theta) \times P(\theta)$$

Step-down: rank p-values smallest to largest; apply sequentially tightened thresholds (α/k , $\alpha/(k-1)$, ...). Less conservative than Bonferroni while still controlling FWER.

FDR (BENJAMINI-HOCHBERG)

Controls expected proportion of false discoveries (not FWER). Less conservative; preferred in exploratory genomic or neuroimaging studies with hundreds of tests.

Example: Post-hoc comparisons after significant ANOVA with 4 groups = 6 pairwise tests. Bonferroni threshold: $p < 0.05/6 = 0.0083$ per comparison.

BAYES FACTOR BF_{10}

Ratio of marginal likelihoods. $BF > 3$: moderate evidence for H_1 . $BF > 10$: strong. $BF < 1/3$: evidence for H_0 . $BF = 1$: no evidence either way. No arbitrary threshold unlike $p < 0.05$.

CREDIBLE INTERVALS

A 95% CrI means there is 95% probability the true parameter lies within it — more intuitive than frequentist confidence intervals (which do not mean this).

BAYESIAN PAIRED T-TEST

Replaces frequentist paired t-test. Reports BF_{10} and posterior of mean difference instead of t and p-value.

Example: Bayesian t-test for CBT vs control on anxiety: $BF_{10} = 12.4 \rightarrow$ strong evidence CBT reduces anxiety. Can also quantify evidence for the null (H_0), unlike NHST which can only reject it.

EFFECT SIZE QUICK REFERENCE TABLE

Test	Effect size measure	Small	Medium	Large	Notes
t-tests (independent & paired)	Cohen's d	< 0.3	0.3–0.5	> 0.5	Lectures: trivial <0.1, small 0.1–0.3. $d = (M_1 - M_2)/SD_{\text{pooled}}$. Large n can produce significant p with trivial d.
ANOVA (between-subjects)	η^2 (eta squared)	0.01	0.06	0.14	$\eta^2 = SS_{\text{between}}/SS_{\text{total}}$. Partial η^2 is more common in factorial designs and reported in SPSS.
Repeated Measures ANOVA	Partial η^2 / ω^2	0.01	0.06	0.14	ω^2 is less biased than η^2 . Report alongside F, df, and p.
Correlation (Pearson, Spearman)	r / ρ	0.10	0.30	0.50	r itself is the effect size. r^2 = shared variance. Report alongside t(df) and p.
Chi-square (2×2)	ϕ (phi)	0.10	0.30	0.50	$\phi = \sqrt{\chi^2/n}$. Same threshold as r. Report: $\chi^2(df, N=n) = x.xx$, $p = .xxx$, $\phi = x.xx$.
Chi-square (>2×2 table)	Cramér's V	0.10	0.30	0.50	$V = \sqrt{\chi^2/n \cdot df^*}$ where $df^* = \min(\text{rows}-1, \text{cols}-1)$. Report alongside χ^2 , df, N, p.
Non-parametric (Mann-Whitney, Wilcoxon, Kruskal)	$r = Z/\sqrt{N}$	0.10	0.30	0.50	Convert Z from test output. Same thresholds as Pearson r. Often overlooked but important to report.
Regression	R^2 (or $f^2 = R^2/(1-R^2)$)	$f^2 = 0.02$	$f^2 = 0.15$	$f^2 = 0.35$	R^2 = total variance explained. Adjusted R^2 penalises extra predictors. f^2 is Cohen's effect size for regression.
Logistic regression	Nagelkerke R^2 / AUC	—	—	—	No universal thresholds. AUC: 0.5 = chance, 0.7–0.8 = acceptable, >0.9 = excellent discrimination.